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EVEN END Semester Examinations MAY 2025

Name of the Course: **B.Tech.,**

Semester: **8th**

Name of the Paper: **Vehicle Body Safety and Control**

Paper Code: **(TAE 801)**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Define active and passive safety in automotive design. Provide examples of each from modern vehicles. (CO1)	CO1, CO4, and CO5
(b)	Identify various types of restraint systems in vehicles. Explain the working of safety belts and airbags. (CO4)	
(c)	Explain human impact tolerance and the concept of Severity Index. How does anthropometry assist in injury prevention? (CO5)	
Q2	(20 marks)	
(a)	Describe different types of crash tests carried out to ensure vehicle occupant safety. What are rebound and roll-over crash tests? (CO2)	CO2, CO5, and CO2
(b)	Discuss the need for pedestrian protection in crash scenarios. Explain how injury criteria are used to determine crash safety. (CO5)	
(c)	How is passenger compartment integrity evaluated during crash testing? What structural safety features are typically assessed? (CO2)	
Q3	(20 marks)	
(a)	Describe the role of ergonomics in vehicle safety system design. How are controls and seating arranged based on ergonomic principles? (CO3)	CO3, CO4, and CO6
(b)	Explain the significance of rearward visibility and safety mirrors in modern automobiles. How are these components tested? (CO4)	
(c)	What are cruise control systems and how do they function? Discuss digital cruise control with throttle actuator integration. (CO6)	
Q4	(20 marks)	
(a)	What is the function of a bumper in impact scenarios? Explain its importance in reducing crash effects. (CO4)	CO4, CO6, and CO6
(b)	Describe the working of an anti-lock braking system (ABS) and electronic suspension system. How do they enhance safety? (CO6)	
(c)	Discuss electronic steering control in modern vehicles. How does it differ from traditional steering systems? (CO6)	
Q5	(20 marks)	
(a)	What is survival space in a vehicle crash context? How is it ensured during vehicle design? (CO1)	CO1, CO2, and CO3
(b)	Explain the relevance of crumple zones and how they contribute to vehicle crashworthiness. (CO2)	
(c)	Discuss the role of seating design in occupant safety. How does ergonomics influence the location of vehicle controls? (CO3)	

